Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Data & Software Engineer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web and mobile applications, building data engineering pipelines, operational analytics assets, and automated transactional workflows in agile sprints using Jira.
- **Order Procurement & Management Solution** | *Python, SQL, PostgreSQL, NextJS, TypeScript*
  - Architected an order procurement platform with PostgreSQL and NextJS, building RESTful APIs to centralize operational process data and automate order workflows for 20+ daily business orders.
  - Designed administrative analytics dashboards and an order management interface supporting employee dispatch, audit logging, and automated reporting to track order fulfillment KPIs.
  - Developed an automated data ingestion pipeline using Python, integrating an LLM agent with RAG fallback and evals to parse unstructured messages into structured SQL records, eliminating manual transactions.
  - Modeled relational PostgreSQL schemas and SQL views with indexing constraints, ensuring data integrity, reducing query latencies by 40%, and delivering clean analytics assets for stakeholders.

Data Scientist & Software Developer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Python, OpenCV, C++, Django, Flutter* **(Live link)**
  - Engineered a native C/C++ bridge using `dart:ffi` to port OpenCV SIFT algorithms into Flutter, enabling cross-platform statistical keypoint extraction across iOS and Android.
  - Built a real-time video data pipeline normalizing camera frame buffers into C structs, maintaining zero memory leaks and 30+ FPS performance during continuous stream transformation.
  - Implemented statistical feature matching models on vector coordinate feeds, validating hypothesis thresholds on spatial objects in Python and C to render real-time AR UI overlays.
  - Constructed RESTful backend APIs in Django and Python, conducting exploratory analysis on spatial location data to deliver location-aware discovery features to 100+ alpha testers.
- **Kia National Conference App** | *Python, Flutter, SQL, Jira*
  - Created an event analytics application for 500+ attendees at the Kia National Dealer Conference, building real-time telemetry tracking and reporting dashboards to analyze user journey metrics.

Technical Skills

Data Science & Analytics: Python, SQL, Pandas, NumPy, Scikit-learn, R, Mathematica, Seaborn, Matplotlib
Foundation Models & AI: RAG, AI Agents, Model Evals, LLM Pipelines, Context Engineering, OpenCV (SIFT)
Databases & Analytic Tools: PostgreSQL, Celonis EMS, Tableau, Power BI, Django, NextJS, NodeJS, ExpressJS
Developer Tools & Agile: Git, Linux, Bash, Docker, AWS, Jira, Agile, LaTeX

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)